AL-Farabi University College
Department of Computer Engineering

Automatic certificate generator system using MATLAB

Research submitted to the AL-Farabi University College, in partial fulfillment of the requirements for a bachelor's degree in Department of ComputerEngineering

By***(Ali Wissam Ali &BadarZaidAbd al-Karim)***

Supervisor

Asst. Lect . Alaa Mohammed

Supervisor certification

I certify that this project titled "*Automatic certificate generation*" by (*Ali Wissam Ali &BadarZaidAbd al-Karim)* was prepared under my supervision in the Department of Computer Engineering and is a partial fulfillment of the requirements for a Bachelor's degree in Computer Engineering.

Signature:

Asst. Lect . Alaa Mohammed

Date: 2022 /

Examination Committee Certification

I certify that have read this project entitled (***automatic certificate generation)***, and as an examining committee examined the students (***Ali Wissam Ali &BadarZaidAbd al-Karim)*** in its contents and in opinion, it meets the standards of the B.Sc. in Computer Engineering.

Signature:
Name:
Scientific Degree:
Date:
(Member)

Signature:
Name:
Scientific Degree:
Date:
(Member)

Signature:
Name:
Scientific Degree:
Date:
(Chairman)

Signature:
Name:
Scientific Degree:
Date:
Head of Computer Engineering

Acknowledgement

I would like to express my deep of gratitude to my supervisor for her patient valuable guidance and encouragement during the supervision of this project.

Asst. lecturer Alaa Mohammed.

My thanks and gratitude are dedicated to the University of Farabi (Computer Engineering Department) for providing the facilities to do this project.

Thanks, are also to my parents encouraging during the period of my study.

Abstract

This paper presents an e-certificate generation system with elegant templatedesigns that contributes to the environment by saving trees and organizationalresources with concise planning and effective management of records. Thesystem is not only designed and developed to save paper but also to organizethe distribution in an effective way to minimize aberration and speed up theprocess 100 times it usually takes with a simple and user friendly designinterface. This research work enables an end-user to choose their desiredcertificate template and template format without any prerequisite knowledge,just by knowing simple Excel sheet.

List of Contents

Acknowledgment i

Abstract..ii

Chapter One INTRODUCTION...1

.1.1 Introduction 2

.1.2 Literature Survey...2

1.3 Aim of Research..4

Chapter Two Software and Tools 5

2.1 MATLAB 6

2.2 Microsoft Excel 10

Chapter Three Practical framework ………………………………..19

3.1 Connection..20

3.2 Code 22

3.3 Conclusion and future work...24

3.4 Results..25

3.5 References...26

List of Figures

Figure1.1……........………………………………………………………..9

Figure1.2………………………………………..……………………..18

Figure1.3…..…………………………………..……………..21

Figure1.4…………………………………………………………25

Chapter One

Introduction

1.1 Introduction

The usage of computer software is widespread. Several organizations perform their daily activities more efficiently using applications founded by the use of computer science knowledge. Certificate Generation System can be used

Certificate Generation System can be used in various universities to automate the distribution of digitally verifiable sheets of students to ease the work. This system verifies the participants' information from an Excel sheet and generates the certificates of all the participants in an image, using MATLAB, it doesn't print the data directly first it checks the number of needed certificates then it provides enough resources to prevent crashes.

1.2 Literature Survey

Hao Wang et Mate Al [1] They offer a secure electronic health record (EHR) system based on special-based Cryptococcus and blockchain technology. This system carried out the ABE as well as IBE to encrypt medical data and to use IBS to apply digital signatures. . In order to obtain various functions of ABI, IBE and IBS in crypto, we present a new cryptographic original; it is called a joint identity-based encryption as well as signature. It simplifies system maintenance and don't require the installation of separate cryptographic system for various security requirements.

In addition, we use blockchain techniques to ensure the integrity and inspection of medical data.

Finally, it is called a joint identity-based encryption as well as signature. It simplifies system maintenance and don’t require the installation of separate cryptographic system for various security requirements.

In addition, we use blockchain techniques to ensure the integrity and inspection of medical data. Finally, we offer a demonstration application for medical insurance business. According to Yan Michalevskyet.

Al [2] system introduces the first practical decentralized ABE scheme with proof of policyhiding. Our creation is based on the basic encryption of decentralized internal product, which is an encryption strategy launched in this paper.

This ABB scheme supports results, disputes, and threshold policies, which protect the access policies of those parties that are not authorized to decrypt content. In addition, we handle the receiver's privacy issue.

Using our plan with Vector Commitment, we hide a complete set of attributes presented by the individual with the recipient; just disclose the feature that regulates the authority. Finally, we propose random-polynomial encoding that immerses this scheme in the presence of corrupt officials.

Al [3] they successfully address these issues by offering a cleared policy feature-based data sharing plan with direct cancellation and keyword search. In the proposed scheme, the non-terminated users' private key is not required to be updated during the cancellation of direct revocation of features. In addition, a keyword search has been realized in our plan, and the search is stable with the increase in time features. Specifically, the policy is hidden in our plan, and therefore, the privacy of users is preserved. Our security and performance analysis show that theproposed plan can deal with security and efficiency concerns in cloud computing.

1.3 Aim of the Research

- To generate certificates automatically with very little manpower, that solution will be conducted using MATLAB.
- To use common tools recognized by the average computer user such as Microsoft excel, so it could be used by nearly anyone.
- To have a flexible program where the user can output any type of images so there will be no image type problems between devices.
- To use a simple code that can be read by any programmer and allow it to be expanded developed.

Chapter Two

Software and Tools

2.1 MATLAB

MATLAB is a high-performance language for technical computing. It integrates computation, visualization, and programming in an easy-to-use environment where problems and solutions are expressed in familiar mathematical notation. Typical uses include:

- Math and computation
- Algorithm development
- Modeling, simulation, and prototyping
- Data analysis, exploration, and visualization
- Scientific and engineering graphics
- Application development, including Graphical User Interface building

MATLAB is an interactive system whose basic data element is an array that does not require dimensioning. This allows you to solve many technical computing problems, especially those with matrix and vector formulations, in a fraction of thetime it would take to write a program in a scalar noninteractive language such as C or Fortran.

The name MATLAB stands for matrix laboratory. MATLAB was originally written to provide easy access to matrix software developed by the LINPACK and EISPACK projects, which together represent the state-of-the-art in software for matrix computation.

MATLAB has evolved over a period of years with input from many users. In university environments, it is the standard instructional tool for introductory and advanced courses in mathematics, engineering, and science. In industry, MATLAB is the tool of choice for high-productivity research, development, and analysis.

MATLAB features a family of application-specific solutions called toolboxes. Very important to most users of MATLAB, toolboxes allow you to *learn* and *apply* specialized technology. Toolboxes are comprehensive collections of MATLAB functions (M-files) that extend the MATLAB

environment to solve particular classes of problems. Areas in which toolboxes are available include signal processing, control systems, neural networks, fuzzy logic, wavelets, simulation, and many others.

The MATLAB System

The MATLAB system consists of five main parts:

The MATLAB language.

This is a high-level matrix/array language with control flow statements, functions, data structures, input/output, and object-oriented programming features. It allows both "programming in the small" to rapidly create quick and dirty throw-away programs, and "programming in the large" to create complete large and complex application programs.

The MATLAB working environment.

This is the set of tools and facilities that you work with as the MATLAB user or programmer. It includes facilities for managing the variables in your workspace and importing and exporting data. It also includes tools for developing, managing, debugging, and profiling M-files, MATLAB's applications.

Handle Graphics.

This is the MATLAB graphics system. It includes high-level commands for two-dimensional and three-dimensional data visualization, image processing, animation, and presentation graphics. It also includes low-level commands that allow you to fully customize the appearance of graphics as well as to build complete Graphical User Interfaces on your MATLAB applications.

The MATLAB mathematical function library.

This is a vast collection of computational algorithms ranging from elementary functions like sum, sine, cosine, and complex arithmetic, to more sophisticated functions like matrix inverse, matrix eigenvalues, Bessel functions, and fast Fourier transforms.

The MATLAB Application Program Interface (API).

This is a library that allows you to write C and Fortran programs that interact with MATLAB. It include facilities for calling routines from MATLAB (dynamic linking), calling MATLAB as a computational engine, and for reading and writing MAT-files.

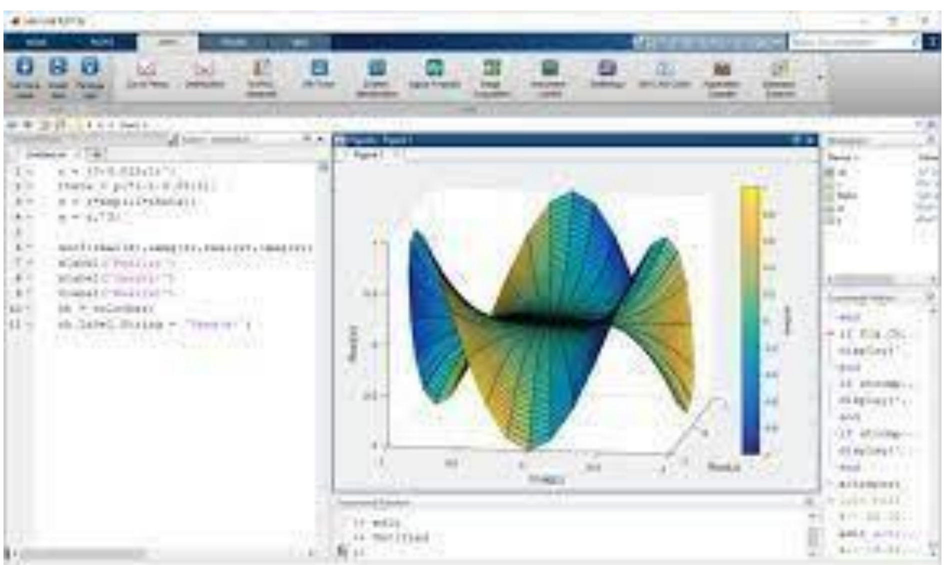

Fig 1.1: Matlab

2.2 Microsoft Excel

Excel is a spreadsheet program from Microsoft and a component of its Office product group for business applications. Microsoft Excel enables users to format, organize and calculate data in a spreadsheet.

By organizing data using software like Excel, data analysts and other users can make information easier to view as data is added or changed. Excel contains a large number of boxes called cells that are ordered in rows and columns. Data is placed in these cells.

Excel is a part of the Microsoft Office andOffice 365 suitesand is compatible with other applications in the Office suite. The spreadsheet software is available for Windows, macOS, Android and iOS platforms.

Common Excel use cases

Excel is most commonly used in business settings. For example, it is used in business analysis,human resource management, operations management and performance reporting. Excel uses a large collection of cells formatted to organize and manipulate data and solve mathematical functions. Users can arrange data in the spreadsheet usinggraphing tools,pivot tablesand formulas. The spreadsheet application also has a macro programming language called Visual Basic for Applications .

Organizations use Microsoft Excel for the following:

- collection and verification of business data;
- business analysis;
- data entry and storage;
- data analysis;
- performance reporting;
- strategic analysis;
- accounting and budgeting;
- administrative and managerial management;
- account management;
- project management; and
- office administration.

Excel terminology and components

Excel has its own terminology for its components, which new users may not immediately find understandable. Some of these terms and components include the following:

- **Cell.** A user enters data into a cell, which is the intersection of a column and row.

- **Cell reference.** This is the set of coordinates where a cell is located. Rows are horizontal and numbered whereas columns are vertical and assigned a letter.
- **Active cell.** This is the currently selected cell, outlined by a green box.
- **Workbook.** This is an Excel file that contains one or more worksheets.
- **Worksheet.** These are the different documents nested within a Workbook.
- **Worksheet tab.** These are the tabs at the bottom left of the spreadsheet.
- **Column and row headings.** These are the numbered and lettered cells located just outside of the columns and rows. Selecting a header highlights the entire row or column.
- **Formula.** Formulas are mathematical equations, cell references or functions that can be placed inside a cell to produce a value. Formulas must start with an equal "=" sign.

- **Formula bar.** This is the long input bar that is used to enter values or formulas in cells. It is located at the top of the worksheet, next to the "fx" label.
- **Address bar.** This bar located to the left of the formula bar shows the number and letter coordinates of an active cell.
- **Filter.** These are rules a user can employ to select what rows in a worksheet to display. This option is located on the top right of the home bar under "Sort & Filter." An auto filter option can be selected to show rows that match specific values.
- **AutoFill.** This feature enables users to copy data to more than one cell automatically. With two or more cells in a series, a user can select both cells and drag the bottom right corner down to autofill the rest of the cells.
- **AutoSum.** This feature enables users to add multiple values. Users can select the cells they want to add and press the Alt and Equal keys. There is also a button to enable this feature on the top right of the home page, above "Fill" and to the left of "Sort & Filter."
- **PivotTable.** This data summarization tool sorts and calculates data automatically. This is located under the insert tab on the far left.

- **PivotChart.** This chart acts as a visual aid to the PivotTable, providing graph representations of the data. It is located under the middle of the insert page, next to maps.
- **Source data.** This is the information that is used to create a PivotTable.

Advanced Excel capabilities

- More advanced tools in Excel include the following:
- **TREND function.** This tool is used to calculate linear trend lines through a set of Y or X values. It can be used for time series trend analysis or projecting future trends. Trendlines can be used on charts.
- **VLOOKUP.** The Vertical Lookup, or VLOOKUP function, can be used to search for values in a larger data set and pull that data into a new table. VLOOKUP is a cell input command that looks like =VLOOKUP(). The
- parentheses include the data the user wants to look up, where to look for it, the column number with the value to return; or optionally, the user can specify an Approximate or Exact match indicated by True or False.
- **Table Array.** This is a combination of two or more tables with data and values linked and related to one another. This is part of VLOOKUP.
- **Col_index_num.** Another value when creating a table array that specifies the column from where data is being pulled.

- **Range_lookup.** This value in VLOOKUP provides information closest to what a user wants to find when nothing matches other variables. This is represented by a true or false label. False gives the exact value a user is looking for and True gives results from a variable data range.
- **MAX and MIN functions.** These functions provide the maximum and minimum values from selected data sets. MAX is used to find the maximum value in a function tab and MIN is used to find the minimum value.
- **AND function.** This function has more than one criteria set when searching variables. If a variable matches the criteria, the value will be returned as true; if not, it will be returned as false. The input for the function should look like this: =AND (logical1, [logical2], ...).

Additional functions for use in Excel include subtract, multiply, divide, count, median, concatenate and otherlogical functionssimilar to AND, such as OR.

Excel and XLS files

An XLS file is a spreadsheet file that can be created by Excel or other spreadsheet programs. The file type represents an Excel Binary File format. An XLS file stores data as binary streams -- a compound file. Streams and substreams in the file contain information about the content and structure of an Excel workbook.

Versions of Excel after Excel 2007 use XLSX files by default, since it is a more open and structured format. Later versions of Excel still support the creation and reading of XLS files, however. Workbook data can also be exported in formats including PDF, TXT, Hypertext markup language, XPS and XLSX.

Macro-enabled Excel files use the XLSM file extension. In this case,macrosare sets of instructions that automate Excel processes. XLSM files are similar to XLM files but are based on the Open XML format found in later Microsoft Office software.

Excel competitors

Even though Excel might be one of the most recognizable spreadsheet programs, other vendors offer competing products. Examples include the following:

- **Google Sheets.**Google Sheets is a free competitor to Excel, with similar layouts and features. Users with a Gmail account can access Google Sheets. Google Sheets are saved in the cloud, meaning users can access their spreadsheets from anywhere and on numerous devices. Multiple users can also collaborate on the same spreadsheet.

- **Numbers.** Apple's spreadsheet program comes free with every Mac and provides prebuilt templates, charts and graphs. Numbers excels at graphics and charts, but it does not handle large data sets as well as Microsoft Excel. Numbers is also exclusive for Apple's devices. But it does enable users to save spreadsheets as Excel files, so a Windows user can still open a Numbers spreadsheet in Excel.
- **Apache OpenOffice Calc.** This free open source spreadsheet software features multiple user collaboration; natural language formulas that enable users to createformulas using words; DataPilot, which pulls data from corporate databases; and style and formatting features that enable different cell formatting options. The software uses a different macro programming language than Excel and has fewer chart options. OpenOfficeCalc works on Windows and macOS platforms. OpenOfficeCalc also uses the Open Document Format as its default, with only limited support for Microsoft's XLSX format.

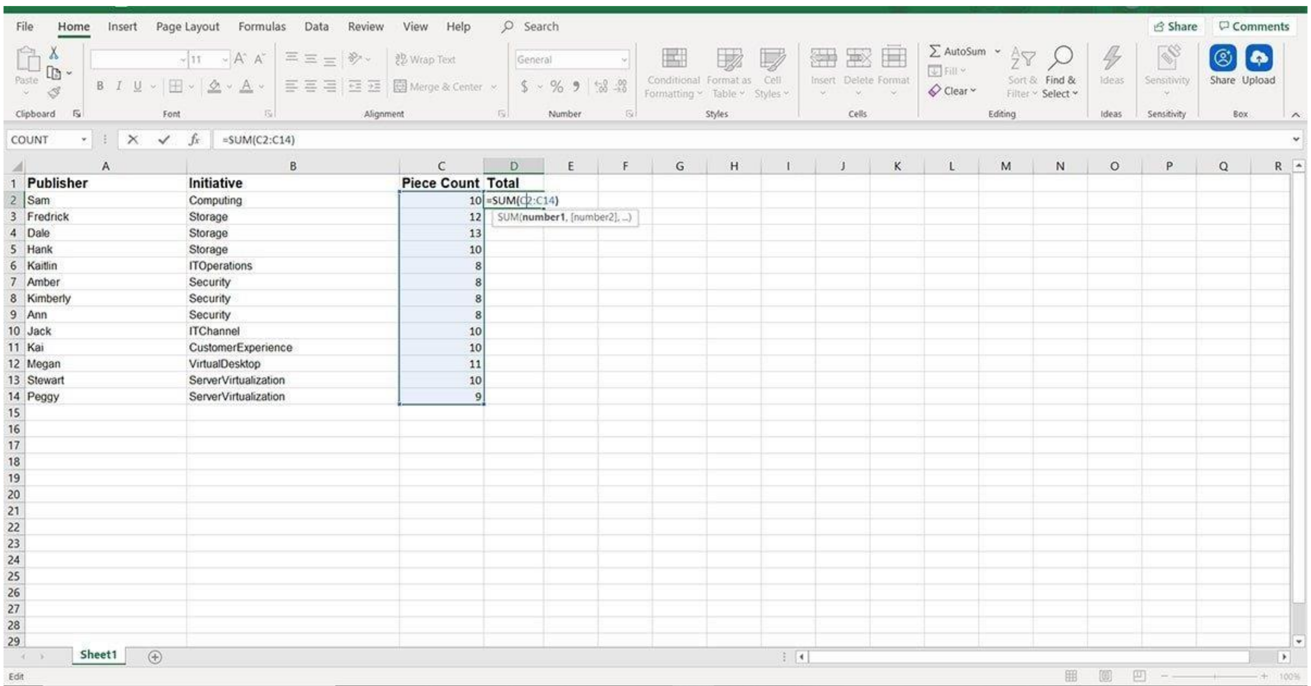

Fig 1.2: Microsoft Excel

Chapter Three

Practical framework

3.1 connection

1. First, add the details of the course participants in an Excel sheet.
2. A base image that is edited or modified by an image editing software that is ready to be used by Matlab any type of images are accepted for it can be changed from Matlab.
3. Add the details of participants in the Excel sheet the details will use the base image to produce different image certificates.
4. The program in Matlab will obtain text length from text length for scalability purposes And number of certificate to be generated so it can handle the amount of Work needed and provide resources for it without crashing the program.

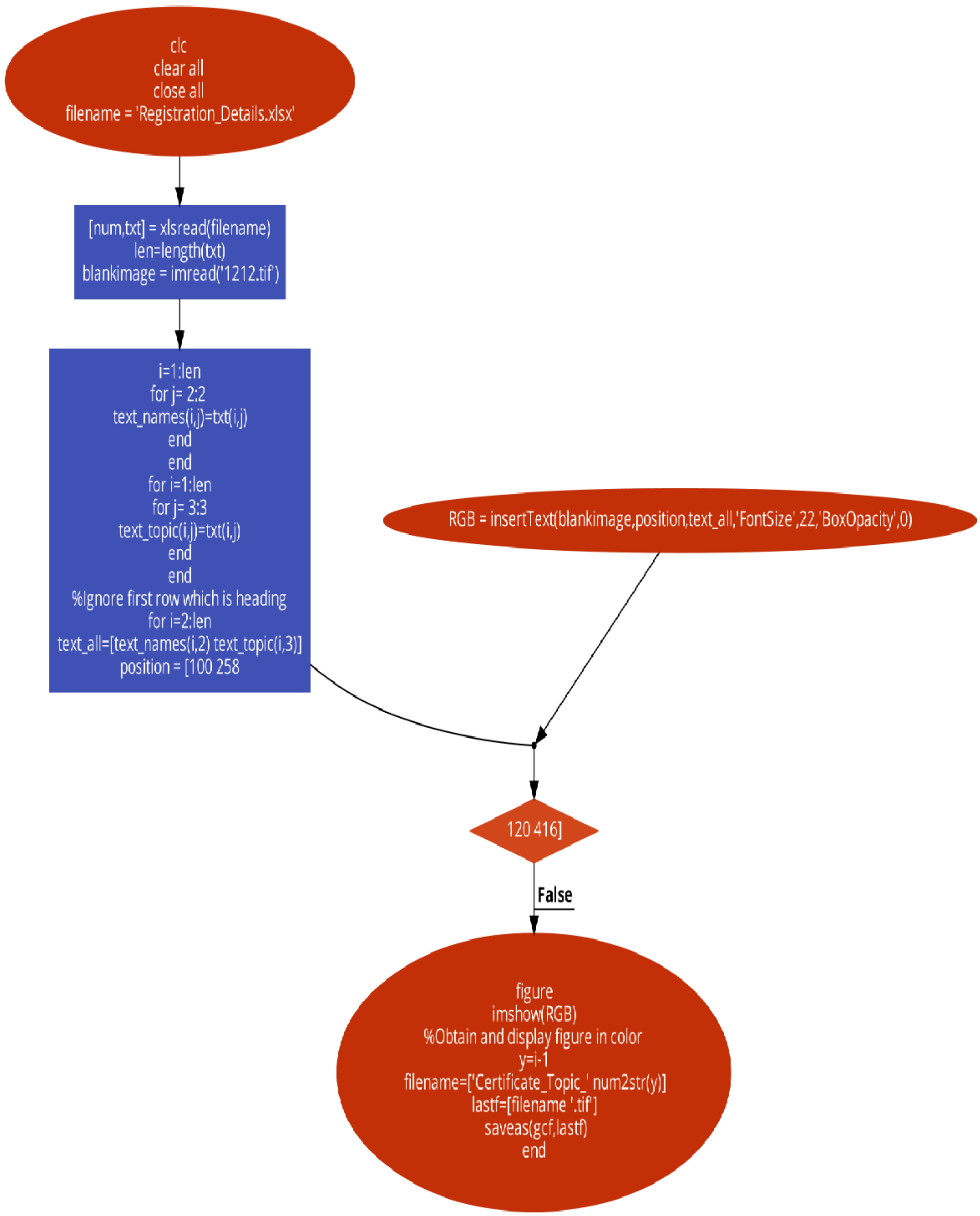

Fig 1.3: Program flow chart

3.2 Code

```
clc% Clear command window.
clear all%Clear variables and functions from memory close all% closes all the open figure windows
home % Send the cursor home

filename = 'Registration_Details.xlsx';

[num,txt] = xlsread(filename)
% Read Excel sheet containing details. Text is read from the file
% seperately from numbers



len=length(txt)
% obtain length of text in excel or number of certificates to be generated
% This code provides scalability


blankimage = imread('1212.tif');
% Read blank certificate image


for i=1:len     for j= 2:2
text_names(i,j)=txt(i,j)     endend
% Obtain names from the txt variable which are in 2nd column


for i=1:len     for j= 3:3      text_topic(i,j)=txt(i,j)     endend
% Obtain topics from the txt variable which are in 3rd column
```

```
%Ignore first row which is heading
for i=2:len
text_all=[text_names(i,2) text_topic(i,3)]
% combine names and topics


    position = [100 258;120 416];
% obtain positions to insert on image, MSPaint or any image editor


    RGB = insertText(blankimage,position,text_all,'FontSize',22,'BoxOpacity',0);
%Provide parameters for function insertText
%Font size is 22 and opacity of box is 0 means 100% transparent


    figure
imshow(RGB)
%Obtain and display figure in color
        y=i-1
    filename=['Certificate_Topic_' num2str(y)]
lastf=[filename '.tif']        saveas(gcf,lastf)
% generate and save files with .tif extensionend


% Code works for any length of data ROWWISE AND COLUMNWISE
% Last for loop with minor changes facilitate new project implemntations
```

3.3 Conclusions and Future Work

We have generated acertificate generation system which, gets the information regarding the event such as event name, organization name, organizing authorities, and their designations by the user. All this essential information with proper algorithm design and architecture leads to the successful result of the enlivening certificate to appreciate participants and students. The system saves on paper, cuts management costs, prevents document forgery, and provides accurate and reliable information on digital certificates.

To work with large data set and multiple data nodes with will be the interesting work in future direction.

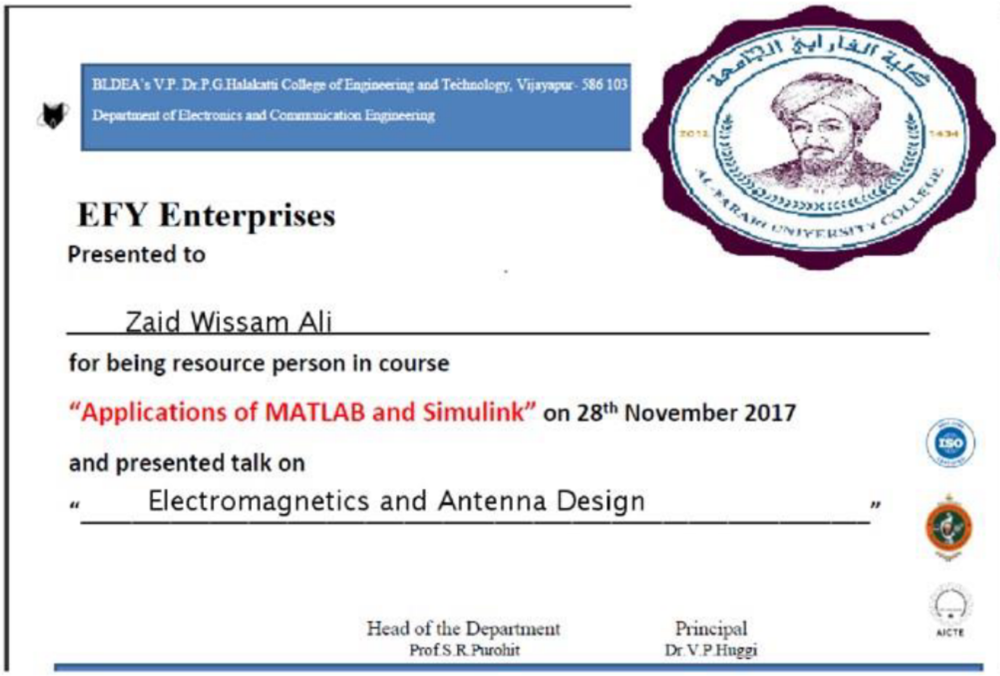

Figure 1.4: Result example

3.4 Results

As we see in figure 1.4 the certification is presented for Zaid Wissam Ali filling the plain space, and Electromagnetic magnetic and Antenna Design filling the second plain space, the right top corner representing the logo of the organization and in the bottom the needed signatures, everything is interchangeable in the image so the user can change anything even the whole image with the consideration of the coordinates.

3.5 References

1. Wang, Hao, and Yujiao Song. "Secure cloud-based EHR system using attribute-based cryptosystem and blockchain." Journal of medical systems 42.8 (2018): 152

2. Michalevsky Y, Joye M. Decentralized Policy-Hiding Attribute-Based Encryption with Receiver Privacy.

3. Wu, Axin, et al. "Hidden policy attribute-based data sharing with direct revocation and keyword search in cloud computing." Sensors 18.7 (2018): 2158

4. SamitShivadekar, Stephen Raj Abraham, Sheikh Khalid, "Document Validation and Verification System", International Journal of Advanced Research in Computer Engineering & Technology (IJARCET), Volume 5, Issue 3, March 2016.

5. NeethuGopal, Vani V Prakash, "Survey on Blockchain-Based Digital Certificate System", International Research Journal of Engineering and Technology (IRJET), Volume 5, Issue 11, Nov. 2018. [6] KiranGautam and DivyaUpadhyay, “Implementing Dynamic Certificates for Securing Database,” in IEEE 5th International Conference, 25-26 Sept. 2014.

6. “Smart Contracts,” http://searchcompliance.techtarget.com/definition/ smartcontract, 2017, [Online; accessed 4-Dec- 2017]

7. Dorri, S. S. Kanhere, and R. Jurdak, “Blockchainin internet of things: Challenges and Solutions,”arXiv:1608.05187 [cs], 2016. [Online]. Available: http://arxiv.org/abs/1608.05187%5Cn http://www.arxiv.org/pdf/1608.05187.pd

8. MathWorks Real-Time Workshop. http://www.mathworks.com/products/rtw.

9. RTCA Special Committee 167. Software considerations in airborne systems and equipment certification. Technical report, RTCA, Inc., December 1992. [10] Ingo Stürmer and Mirko Conrad. Test suite design for code generation tools. In Proceedings of 18th IEEE International Conference on Automated Software Engineering}, pages 286-290. IEEE, October 2003. [11] Ingo Stürmer, Daniela Weinberg, and Mirko Conrad. Overview of existing safeguarding techniques for

الخلاصة

تقدم هذه الورقة نظامًا لإنشاء الشهادات الإلكترونية مع تصميمات قوالب أنيقة تساهم في البيئة من خلال توفير الأشجار والموارد التنظيمية من خلال التخطيط المختصر والإدارة الفعالة للسجلات. لم يتم تصميم النظام وتطويره لتوفير الورق فحسب ، بل أيضًا لتنظيم التوزيع بطريقة فعالة لتقليل الانحراف وتسريع العملية 100 مرة التي تستغرقها عادةً مع واجهة تصميم بسيطة وسهلة الاستخدام. يمكّن هذا العمل البحثي المستخدم النهائي من اختيار قالب الشهادة المطلوب وتنسيق القالب دونمعرفة مسبقة ، فقط من خلال معرفة ورقة اكسل البسيطة.

كلية الفارابي الجامعة
قسم هندسة الحاسوب

نظام مولد الشهادات الأوتوماتيكي باستخدام MATLAB

مشروع التخرج مقدم الى قسم هندسة الحاسوب كجزء من متطلبات نيل درجة

البكالوريوس في هندسة الحاسوب

من قبل علي وسام علي و بدر زيد عبد الكريم

المشرف

مساعد مدرس : الاء محمد